SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 1 **Total Marks:10**

Annexure A

Descriptive Written Test

Code of Conduct:

I RAHIMA BANU N-192525009 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: RAHIMA BANU N

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation Using Intelligent Agents

1. Introduction

A smart home automation system uses Artificial Intelligence (AI) to control lighting, temperature, and security based on user behavior and environmental conditions. It receives real-time inputs from motion sensors, temperature sensors, cameras, and time-based schedules. The system learns user preferences and automatically adjusts home settings to improve comfort, energy efficiency, and security.

Intelligent agents help the system perceive its environment, make decisions, and perform appropriate actions.

2. Types of Intelligent Agents

1. Simple Reflex Agent

A simple reflex agent selects actions based on the current input using predefined condition-action rules. It does not consider past experiences or maintain an internal model of the environment.

Function in a smart home:

- Turns on lights when motion is detected.
- Turns off lights when no motion is detected for a specified period.
- Activates an alarm when a door sensor detects unauthorized entry.

Example: If motion is detected in a room, the agent switches on the lights.

Advantages: Simple, fast, and easy to implement.

Limitations: Cannot learn user preferences or make decisions based on past conditions. It may turn off lights even when a user is sitting still.

2. Model-Based Reflex Agent

A model-based reflex agent maintains an internal representation of the environment. It uses current sensor readings and previous information to understand the situation and make decisions.

Function in a smart home:

- Tracks room occupancy using motion sensors and previous sensor readings.
- Maintains the current temperature and heating or cooling status.
- Determines whether a room is occupied before switching off lights or air conditioning.
- Uses door and window sensor states to improve security decisions.

Example: If motion is temporarily absent but the system knows the room is occupied, it keeps the lights on.

Advantages: Handles partially observable environments and makes more context-aware decisions.

Limitations: Requires an accurate internal model and may not independently determine the best long-term outcome.

3. Goal-Based Agent

A goal-based agent selects actions that help achieve specific objectives. It evaluates possible actions and chooses those that move the system toward its goals.

Function in a smart home:

- Maintains a comfortable room temperature.
- Ensures that doors are locked when the home is unoccupied.
- Turns off unnecessary electrical appliances.
- Adjusts lighting according to user schedules and activities.

Example: If the goal is to maintain a temperature of 24°C, the agent activates the air conditioner when the room temperature rises above the target.

Advantages: Flexible decision-making and the ability to plan actions to achieve desired outcomes.

Limitations: May not efficiently compare trade-offs between comfort, energy consumption, and security when several goals conflict.

4. Utility-Based Agent

A utility-based agent evaluates possible actions using a utility function that measures how desirable each outcome is. It selects the action that provides the highest expected overall benefit.

Function in a smart home:

- Balances comfortable temperature against electricity consumption.
- Adjusts lighting based on occupancy and available daylight.
- Prioritizes security while minimizing unnecessary alarms.
- Learns user preferences from repeated behavior and adjusts its decisions.

Example: If the user prefers a temperature of 24°C but energy prices are high, the agent may choose a slightly different temperature within the user's comfort range to reduce electricity costs.

Advantages: Handles conflicting objectives, supports optimization, and can make decisions that balance multiple requirements.

Limitations: Requires a well-designed utility function and reliable data. Incorrect priorities can lead to undesirable decisions.

3. Comparison of Intelligent Agents

Agent Type	Decision Basis	Smart Home Application	Main Limitation
Simple Reflex	Current sensor input	Motion-based lighting	No memory or learning
Model-Based	Current input and internal state	Occupancy and temperature tracking	Depends on model accuracy
Goal-Based	Desired objectives	Maintaining comfort and securing the home	Limited trade-off evaluation
Utility-Based	Overall benefit or utility	Balancing comfort, cost, and security	More complex to design

4. Recommended Approach: Hybrid Intelligent Agent

A hybrid intelligent agent combining model-based, goal-based, and utility-based decision-making with simple reflex rules is the most effective approach for a smart home automation system.

The hybrid approach combines the strengths of all four agents.

1. Simple reflex component: Responds immediately to critical events, such as triggering an alarm when a verified intrusion signal is received.
2. Model-based component: Maintains information about occupancy, temperature, door status, and recent sensor activity.
3. Goal-based component: Works toward objectives such as maintaining a comfortable temperature and ensuring that the home is secured.
4. Utility-based component: Balances comfort, energy efficiency, operating costs, and security when selecting the best action.
5. Learning component: Uses user feedback and repeated behavior to personalize lighting schedules and preferred temperature settings.

Example of Hybrid Agent Operation

Suppose the user leaves home during the afternoon.

- Motion and door sensors indicate that the home is unoccupied.
- The model-based component updates the occupancy status.
- The goal-based component identifies the objectives of saving energy and securing the house.
- The utility-based component determines suitable temperature and lighting settings while preserving security.
- The reflex component responds immediately to critical sensor events.
- The learning component improves future settings based on the user's preferences.

5. Expected Outcomes

1. Improved comfort: Automatic adjustment of lighting and temperature according to user preferences.
2. Energy efficiency: Reduced unnecessary use of lights, air conditioning, and other appliances.
3. Enhanced security: Faster responses to suspicious activity and improved monitoring.
4. Personalization: Better adaptation to user routines over time.
5. Real-time responsiveness: Immediate action for urgent events and sensor changes.

6. Conclusion

Simple reflex, model-based, goal-based, and utility-based agents provide different levels of intelligence for smart home automation. Simple reflex agents are suitable for immediate responses, model-based agents maintain environmental context, and goal-based agents work toward specified objectives. Utility-based agents are particularly useful for balancing competing requirements.

Therefore, a hybrid intelligent agent with model-based awareness, goal-based planning, utility-based optimization, reflex responses, and preference learning is the most effective solution. It provides a practical balance of comfort, energy efficiency, security, and personalized automation.